

Dr Alina Catalina Donea : Curriculum Vitae
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Senior Lecturer in Applied Mathematics and Astrophysics
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EDUCATION:

Graduate Certificate of Teaching in Higher Education, 2006, Monash University
PhD in Mathematics/Astronomy, 1999, Romanian Academy & cotutelle Max Planck Institute of Radio-astronomy, Bonn, Germany
Diploma in Physics, 1992, Bucharest University, Physics
Bachelor of Physics, 1992, Bucharest University, Physics

EMPLOYMENT HISTORY:

2008 - Senior Lecturer, Lev C, School of Mathematical Sciences, Monash University ,
2005- 2007 Lecturer B, School of Mathematical Sciences, Monash University ,
2004 Research Fellow, Australian Research Council , Monash University ,
2001- 2003, Postdoctoral Research fellow, Australian Research Council, Adelaide University,
1999 - 2000, Scientist, Astronomical Institute of the Romanian Academy

RESEARCH INTERESTS:

Solar physics: wave oscillations, helioseismic holography, space weather, wave dynamics, acoustic and magnetic waves; sunspots and the quiet sun; solar quakes, solar flares, chromosphere and photosphere hydrodynamics. (observations and interpretation, modelling), magneto-seismology and AI for space weather, mathematical modelling

Mathematical methods, Imaging Tools: cGAN Algorithms for AI, networks, Fourier analysis, Numerical Integration Methods, Astro Data Visualization, Advanced statistics, Laplace methods, Convolutions, Filtering, 1D and 2-D Gaussian Noise, Fourier Transforms, Frequency Spectra, Periodograms, Time series. Statistics of Noise

Increased Impact for future funding: Australian Space Science Networking: My successful novel research of farside helioseismology in 2019 and 2020 and the novel contribution to the development of Magnetic Mapping of Farside Active Regions using artificial Intelligence methods for Space Weather, have led to an increased recognition at national level of my role in the development of a strategic road for the space science in Australia.

OUTREACH

Monash Mathematician to join renowned physicist Professor Brian Cox on the ABC's Stargazing Live Series

Linkedin: <https://www.linkedin.com/in/alina-donea-85a884b0/>

International Women's Day March 2025

Sollar and Stellar quakes, Sydney Skywatcher

Poster: Towards an Australian Centre of Excellence in Space Weather

Monash experts: Forecasting the return of Aurora australis to Victoria

IMPACT:

2022: Guest editor of a 48page special edition of the Australian Physics Magazine "Solar physics and Space Weather", vol 59,2

2018: Co-author of the manual for "The Stanford Synoptic Seismic Monitor of the Sun's Far Hemisphere" in operation at the Joint Science Operations Center Stanford for NASA & Solar Dynamics Observatory. The operating monitor has been supported by 5 NASA contracts, 2 NOAA contracts to NorthWest Research Associates with subcontracts to the Global Oscillations Network Group of the National Solar Observatory.

2022: GitHub public repository Initiative: Author of accessible material for training on machine learning algorithms and Python for visualisation & mathematics, satellite data analysis

2022: Host of a series of hybrid inter/national popular AI workshops 2022+ which resonated with the space related Australian communities, led to interactive sessions, providing a framework of AI resources and collaborations, nationally recognised as essential for the research field.

2023: YouTube channel: Deep Learning For Space Research at MonashU, support platform for AI information

RESEARCH HIGHLIGHTS:

June 2025: Dr Donea's team of international researchers has discovered that sunspots, previously thought to have uniform magnetic fields, are actually fragmented just beneath their surface (Broock et al, 2025). Using data from NASA's Solar Dynamics Observatory, they identified structures called "Strong Acoustic Scatterers" which can only be detected

through helioseismology—the study of solar sound waves. This finding supports a theory from 1979 suggesting that such fragmentation is essential for sunspot stability and may play a role in solar activity like flares. The discovery is considered a significant advancement in understanding the Sun's internal dynamics. Monash University is planning an educational outreach event to highlight this discovery, and discussions are underway for a coordinated press release with NASA to coincide with the publication.

The primary research interests lie in solar wave mechanics and helioseismology, incorporating machine learning. A key challenge in the data science community is the lack of simulated data for testing, and my work directly addresses this by generating large-scale datasets and mathematical visualisation methods. She developed a custom adaptation of helioseismic holography, enabling its successful application to transient emissions from solar flares—a mathematical method that maps the acoustic “loudness” of the Sun. This led to pioneering work in flare helioseismology, quiet Sun acoustics, and multi-wavelength analysis of solar quakes. Her group is the only research group in Australia which successfully adapted Machine Learning algorithms for solar near&far-side images to construct magnetic maps necessary for space weather forecasting.

Flare-seismology(1999–present).A major discovery in 1999 revealed that solar flares generate powerful seismic sources “solar quakes” which launched an entirely new research field now known as *flare seismology*. Using acoustic holography, we imaged the sources of seismic waves driven by a single X-class flare. The signature identified by holography was striking and prompted the recognition of transient seismic emission from dozens of far weaker flares, vastly broadening the scope of the field. Acoustically active flares remain the most compact and transient sources of acoustic emission known on the Sun. Collaborative work with C. Lindsey and D. Braun in Boulder, and with researchers at UC Berkeley, the National University of Colombia (Bogotá), and Monash University has recently led to the discovery of transient acoustic emission originating 1,000–2,000 km beneath the solar surface—well below the flare itself. This result extends flare seismology into the domain of *sub-photospheric solar-interior acoustics*, revealing that flare-driven impulses can penetrate and excite deeper layers of the Sun than previously recognised.

RECENT INVITED PRESENTATIONS AT INTERNATIONAL CONFERENCES

[2025] invited speaker International Astronomical Union Symposium 400 “Solar and Stellar Multi-scale Activity, July Colombia

[2025] Plenary Speaker at the forthcoming AIP Summer Meeting to be held in Wollongong on 1-5 December 2025

[2020] invited talk, 5th Asia Oceania Solar Physics meeting, Pune India, Feb 2020.

[2019] Plenary speaker “Australian Space Agency and Space Research”

[2018] invited speaker on the topic: "Waves and solar flare seismology from photosphere to corona", 2nd Asia- Pacific Conference on Plasma Physics, Kanazawa, Japan, 2018

[2016] invited talk in the STSP sessions at the Australian Institute of Physics Congress (joint with the Plasma Physics Division of the Association of Asia Pacific Physical Societies [AAPPS]) 2016. declines, overlap with the NASA reviewer meeting in Washington DC.

VISITING APPOINTMENTS and Fellowships:

- 2025, North West Research Associates , Boulder
- 2024, University of South Queensland; 2024 Sheffield University
- 2018,2023, High Altitude Observatory & NorthWestResearchAssociates, National Center Atmospheric Research (NCAR), Colorado(fully funded) 2016, Visiting Scientist, High Altitude Observatory, Colorado

SELECTED GRANTS/AWARDS:

2025 Monash High Altitude Balloon, Launching a Telescope , cumulative \$15k

2022-2023 UK Royal Society grant, Collaboration with Sheffield Univ., groundbreaking research on detection of flux emergences before their appearance on the solar surface, \$10k

2024 - OSP Monash Grant \$5.5k

2024 travel support money from NASA Grant 80NSSC20K0712. \$3k

2020-2023 Monash Data Futures Institute Seed Grants, AI & Data Science Monash Global Challenges, \$50k

2013-2014, 2020-2024 NASA fully funded research visit to NWRA & NSO

2018 Monash Advancing Women Research Success Grant

Quality research supervision and mentoring: (2020-2023) Guided & developed the capabilities of four research assistants by engaging them in AI research projects transferred to industry applications (2008-) Supervised as main and associate supervisor: 13 PhD students (3 active), over 16 Honours research projects.

RECENT LEADING ROLES:

Leader and strategic director of the Machine Learning and Data Applications Group in Solar Physics at Monash University, supported by the Monash Data Futures Institute (MDFI). Your role encompasses pioneering research at the intersection of solar physics and machine learning, focusing on areas such as helioseismology, solar flare acoustics, and space weather forecasting. Under my leadership, the group has organized 3 significant AI workshops, including the "1st,2nd,3rd Workshops on Deep Learning Algorithms for

Space Research and Data Imaging” (2023,2023,2024) aimed at advancing solar physics and space weather research through machine learning techniques. The work has been instrumental in integrating AI methodologies into solar physics

Director of Service Teaching of Mathematics for Engineering and IT 2021 – 2024, Deputy Director of Mathematical Engineering Teaching 2019 –2020

School of Mathematics, member of Education Committee 2017-

Deputy Director, Monash Centre of Astrophysics, 2014-2016; treasurer (2013-2016)

CURRENT PROFESIONAL SERVICE AND ACTIVITIES

2021- Chair of the Solar Terrestrial and Space Physics Group, Australian Institute of Physics

Peer-review activities: **NASA** Panel reviewer of scientific proposals, Washington quarters (2008, 2011, 2016)

Member of: Astronomical Society of Australia, Australian National Institute for Theoretical Astrophysics, the International Astronomical Union, American Astronomical Society : Solar Physics Division

2019 Member of the “Solar Physics” Advisory Board

2014- Editorial Board Member of "Journal Frontiers in Astronomy and Space Sciences"

2000-Editor for Earth and Space Science

2004: Scientific referee for the Astrophysical Journal, Astronomy and Astrophysics, Solar Physics

Conferences: Major role in two Local Organizing Committees and Member of Scientific Organizing Committees for **seven** international conferences

2003- Referee for PhD and Master theses

MEMBERSHIP OF PROFESSIONAL SOCIETIES:

International Astronomical Union (**IAU**), Solar Physics Division (**SPD**) of American Astronomical Society (**AAS**), American Geophysical Union (**AGU**), Astronomical Society of Australia (**ASA**),

SELECTED AWARDS:

(2025) Women in Space professional achievement prize

2025- IGNITE Research Initiative fund, Monash University

(2022-23) UK Royal Society grant, Collaboration with Sheffield Univ., ground breaking research on detection of flux emergencies before their appearance on the solar surface

(2013-2014, 2020-2024) NASA fully funded research visit to NWRA & NSO

2014 Kevin Westfold Distinguished Visitor Award, Monash Univ, Essex U

2014,2016,2018 Visitor position High Altitude Observatory,National Center for Atmospheric Research, Boulder

2014,2024 Monash University Out Study Programs

2016 Monash Univ Applied Maths Visitor Grant, Award collaboration Dr Phil JudgeHighAltitude Observatory

2018 Monash Advancing Women Research Success Grant

2021-2023 Monash Data Futures Institute Seed Grants - AI and Data Science for Monash Global Challenges

SUPERVISION 13 PhD students: 2 main, 3 joint and 10 associated supervisions (3 active) **Quality research supervision and mentoring** Guided & developed the capabilities of four research assistants by engaging them in AI research projects transferred to industry applications; 16 Honours research projects.

RECENT INSTITUTIONAL SERVICE

2025- Academic leader and initiator of the Monash High Altitude Balloon student-led initiative

2021- Leader and strategic director for the Monash Machine learning applications and solar physics group, sponsored by Monash Data Futures Institute (MDFI).

Director of Service Teaching for Engineering and IT 2021 – 2024

Deputy Director of Mathematical Engineering Teaching 2019 –2020

School of Mathematics, member of Education Committee 2017-2024

Deputy Director, Monash Centre of Astrophysics, 2014-2016; treasurer of the Monash Centre for Astrophysics (2013-2016)

OTHERS:

- 2014 Recognised through NASA partnerships as a world leader in holography. Collaborator in major proposal "Exploring the Physical Relationship Among Photospheric Magnetic Field Changes, Sunspot Motions & Sunquakes During Solar Eruptions" (2023/24), developed novel techniques to discover seismicity in small solar flares & chromospheric roles in transport of flare energy into photosphere

- 2008, 2011, 2016 Invited as Expert NASA reviewer for three Solar and Heliospheric Physics Supporting Research & Technology Programs.
- 2018, 2023 Key role in a NASA collaboration: development of the theory behind the anisotropy of seismic ripples, discovery of submerged seismic sources
- 2023 special guest dialogue @Monash Conversations on AI & Data Science;
- 2021 Member of the Working Group on Heliosphere science within Australian Academy's National Committee for Space & Radio Science preparing the strategic plan; chair of the Solar and Space Terrestrial Physics Group of the Australian Institute of Physics.
- 2022- Editor of the high impact Journal Earth and Space Science, American Geophysical Union (AGU)
- Member of International Astronomical Union, Solar Physics Division of American Astronomical Society, Astronomical Society of Australia
- 2003 Development of radiation field models in black holes and prediction of neutrinos from M87, highly cited papers

PUBLICATIONS

<http://adsabs.harvard.edu/Donea> 123 publications, 1410 cites, ADS/NASA 49 referred, 74 non-refereed publications, 4 papers with 120+ citations (Google Scholar); h-index 19; I-10 index 27. Citation rates in solar physics are a third of those in astrophysics more broadly, based on top-ranked generalist journals ApJ, MNRAS, A&A. CI Donea's research practice puts PhD students as first authors in papers.

2022: Guest editor of a 48page special edition of the Australian Physics Magazine "Solar physics and Space Weather", vol 59,2

2018: Co-author of the manual for "The Stanford Synoptic Seismic Monitor of the Sun's Far Hemisphere" in operation at the Joint Science Operations Center Stanford for NASA & Solar Dynamics Observatory. The operating monitor has been supported by 5 NASA contracts, 2 NOAA contracts to NorthWest Research Associates with subcontracts to the Global Oscillations Network Group of the National Solar Observatory.

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